## SECR2043 OPERATING SYSTEMS

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|---|---|---|
| Name | : A F I F  S H A Q I R  I R F A N  B I N  A R Q A M | Marks |
| Student ID | : A 2 3 C S 0 2 0 4 | |
| Section | : 0 3 | |

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

## ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

| | | |
|---|---|---|
| 1. | Display your current working directory. | |
| | **Command** | **Output** |
| | pwd | afifshaqir@secr2043:~$ pwd<br>/home/afifshaqir |
| 2. | List all the files and directories in your home directory, including hidden ones. | |
| | **Command** | **Output** |
| | ls -a / | afifshaqir@secr2043:~$ ls -a /<br>. bin boot etc lib lib64 media opt root sbin snap sys usr<br>.. bin.usr-is-merged dev home lib.usr-is-merged lost+found mnt proc run sbin.usr-is-merged srv tmp var |
| 3. | Create a new directory named "os_lab" in your home directory. | |
| | **Command** | **Output** |
| | mkdir os_lib | afifshaqir@secr2043:~$ mkdir os_lab<br>afifshaqir@secr2043:~$ ls |
| 4. | Change your current working directory to "os_lab". | |
| | **Command** | **Output** |
| | cd os_lab | afifshaqir@secr2043:~$ cd os_lab<br>afifshaqir@secr2043:~/os_lab$ to |
| 5. | Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it. | |
| | **Command** | **Output** |
| | Touch hello.txt<br>echo “Hello, world!”>hello.txt | afifshaqir@secr2043:~/os_lab$ touch hello.txt<br>afifshaqir@secr2043:~/os_lab$ echo "Hello, world!">hello.txt |

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| 6. | Display the content of "hello.txt". | |
| | **Command** | **Output** |
| | cat hello.txt | afifshaqir@secr2043:~/os_lab$ cat hello.txt<br>Hello, world! |
| 7. | Copy "hello.txt" to another file named "hello_copy.txt" in the same directory. | |
| | **Command** | **Output** |
| | cp hello.txt hello_copy.txt | afifshaqir@secr2043:~/os_lab$ cp hello.txt hello_copy.txt<br>afifshaqir@secr2043:~/os_lab$ ls -R<br>.:<br>hello.txt  hello_copy.txt<br>afifshaqir@secr2043:~/os_lab$ cat hello_copy.txt<br>Hello, world! |
| 8. | Rename "hello_copy.txt" to "hello_bak.txt". | |
| | **Command** | **Output** |
| | mv hello_copy.txt hello_bak.txt | afifshaqir@secr2043:~/os_lab$ mv hello_copy.txt hello_bak.txt<br>afifshaqir@secr2043:~/os_lab$ ls -R<br>.:<br>hello.txt  hello_bak.txt |
| 9. | Delete "hello.txt". | |
| | **Command** | **Output** |
| | rm hello.txt | afifshaqir@secr2043:~/os_lab$ rm hello.txt<br>afifshaqir@secr2043:~/os_lab$ ls -R<br>.:<br>hello_bak.txt |
| 10. | Display your home directory with additional information, together with subdirectories contents. | |
| | **Command** | **Output** |
| | ls -R ~ | afifshaqir@secr2043:~/os_lab$ ls -R ~<br>/home/afifshaqir:<br>/home/afifshaqir/os_lab:<br>hello_bak.txt |
| | **Total Mark:** | |

## ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

| No. | Question / Command | Output |
|---|---|---|
| 1. | Create an empty file named "hello.txt" in the current working directory. | |
| | **Command** | **Output** |
| | touch hello.txt | afifshaqir@secr2043:~$ touch hello_bak.txt<br>afifshaqir@secr2043:~$ ls<br>hello.txt hello_bak.txt lab linux-lab |
| 2. | Write the text "Hello World" to "hello.txt" using echo. | |
| | **Command** | **Output** |
| | echo “Hello, world!”>hello.txt | afifshaqir@secr2043:~$ echo "Hello,world!">hello. |
| 3. | Append the text "This is a test" to "hello.txt" using echo. | |
| | **Command** | **Output** |
| | echo “This is a test” >> hello.txt | afifshaqir@secr2043:~$ echo "This is a test">>hello.txt |
| 4. | Display the contents of "hello.txt" using cat. | |
| | **Command** | **Output** |
| | cat hello.txt | afifshaqir@secr2043:~$ cat hello.txt<br>Hello,world!<br>This is a test |
| 5. | Create another file named "bye.txt" with the text "Goodbye World" using echo. | |
| | **Command** | **Output** |
| | touch bye.txt<br>echo “Goodbye World” > bye.txt | afifshaqir@secr2043:~$ touch bye.txt<br>afifshaqir@secr2043:~$ ls<br>bye.txt hello.txt hello_bak.txt lab linux-lab<br>afifshaqir@secr2043:~$ echo "Goodbye World">bye.txt |
| 6. | Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat. | |
| | **Command** | **Output** |
| | cat hello.txt bye.txt | afifshaqir@secr2043:~$ cat hello.txt bye.txt<br>Hello,world!<br>This is a test<br>Goodbye World |

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|---|---|---|
| 7. | Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat. | |
| | **Command** | **Output** |
| | cat hello.txt bye.txt > greeting.txt | afifshaqir@secr2043:~$ cat hello.txt bye.txt >greeting.txt |
| 8. | Display the contents of "greeting.txt" using cat. | |
| | **Command** | **Output** |
| | cat greeting.txt | afifshaqir@secr2043:~$ cat greeting.txt<br>Hello,world!<br>This is a test<br>Goodbye World |
| 9. | Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines. | |
| | **Command** | **Output** |
| | nano greeting.txt | afifshaqir@secr2043:~$ nano greeting.txt<br>GNU nano 7.2<br>Hello,afif shaqir !<br>This is a test<br>Goodbye World |
| 10. | Display the modified contents of "greeting.txt" using cat. | |
| | **Command** | **Output** |
| | cat greeting.txt | afifshaqir@secr2043:~$ cat greeting.txt<br>Hello,afif shaqir !<br>This is a test<br>Goodbye World |
| | **Total Mark:** | |

## ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc.
For each question, write down the command and the output (if any) in the answer sheet.

| | | |
|---|---|---|
| 1. | Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c". | |
| | **Command** | **Output** |
| | nano hello.c | afifshaqir@secr2043:~$ nano hello.c |
| 2. | Compile the program "hello.c" using gcc and generate an executable file named "hello". | |
| | **Command** | **Output** |
| | gcc hello.c -o hello | afifshaqir@secr2043:~/workspace$ gcc hello_world.c -o hello_world |
| 3. | Run the executable file "hello" and display its output. | |
| | **Command** | **Output** |
| | ./hello | afifshaqir@secr2043:~/workspace$ ./hello_world<br>Hello, World! |
| 4. | Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c". | |
| | **Command** | **Output** |
| | nano sum.c | afifshaqir@secr2043:~/workspace$ gcc sum.c -o sum |
| 5. | Compile the program "sum.c" using gcc and generate an executable file named "sum". | |
| | **Command** | **Output** |
| | gcc sum.c -o sum | afifshaqir@secr2043:~/workspace$ gcc sum.c -o sum |
| 6. | Run the executable file "sum" with 10 and 20 as arguments and display its output. | |
| | **Command** | **Output** |
| | ./sum 10 20 | afifshaqir@secr2043:~/workspace$ ./sum 10 20<br>Sum: 30 |
| 7. | Run the executable file "sum" with -5 and 15 as arguments and display its output. | |
| | **Command** | **Output** |
| | ./sum -5 15 | afifshaqir@secr2043:~/workspace$ ./sum -5 15<br>Sum: 10 |

<table>
<tr><td>8.</td><td colspan="2">Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>nano echo.c</td><td><code>afifshaqir@secr2043:~/workspace$ nano echo.c</code></td></tr>
<tr><td>9.</td><td colspan="2">Compile the program "echo.c" using gcc and generate an executable file named "echo".</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>gcc echo.c -o echo</td><td><code>afifshaqir@secr2043:~/workspace$ gcc echo.c -o echo</code></td></tr>
<tr><td>10.</td><td colspan="2">Run the executable file "echo", type "Hello World" as input and display its output.</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>./echo</td><td><code>afifshaqir@secr2043:~/workspace$ ./echo</code><br><code>Enter a line of text: Hello World</code><br><code>You entered: Hello World</code></td></tr>
<tr><td></td><td>Total Mark:</td><td></td></tr>
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